

Week 1

0. Introduction and Orientation

Dr. Andreas S. Maniatis

Assistant Professor

School of Computer Science

COMP.2800 – Software Development [2026W]

Monday, January 5th, 2026 | 16:00 – 17:20 | ER 1118



University of Windsor

A close-up photograph of a white bird's head, likely a grebe, with its long neck and eye visible. The bird is positioned on the right side of the frame, facing left. The background is a solid, bright yellow. The bird's feathers are white and appear soft and downy.

Land Acknowledgment

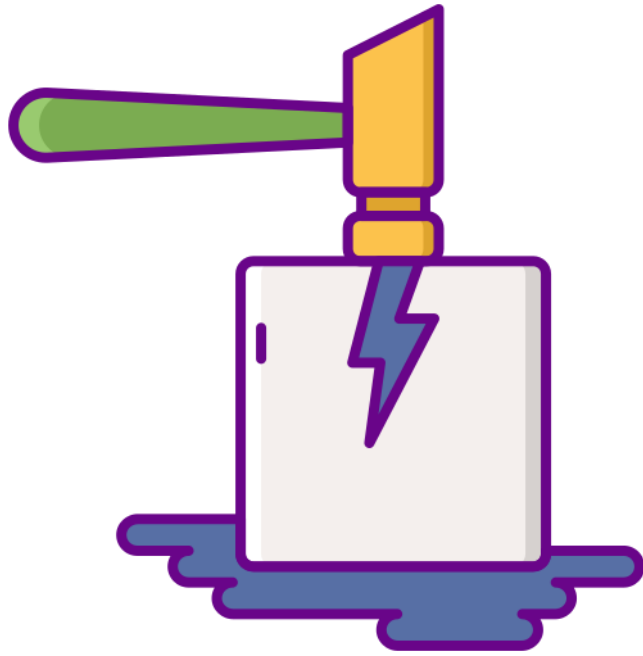
The University of Windsor sits on the traditional territory of the Three Fires Confederacy of First Nations, which includes the Ojibwa, the Odawa, and the Potawatomi.

We respect the longstanding relationships with First Nations people in this place in the 100-mile Windsor-Essex peninsula and the straits – les détroits – of Detroit.



Ice Breaker

How many languages do
you speak?



καλησπέρα • (kalispéra)

你好

HALLO

안녕

HOLA

नमस्ते

CIAO

γεια

HELLO

こんにちは

привет

BONJOUR

مرحبا

OLÁ



University of Windsor

What country do you come from?

- Use the following Padlet:
<https://padlet.com/andreasmaniatis3/where-are-you-from-ci99046lv3yrhvo5>





Andreas Maniatis He/Him

Professor, Big Data Analytics and Data Science | Computer & Software Engineer | Speaker

Windsor, Ontario, Canada • [Contact info](#)

500+ connections



University of Windsor



National Technical University
of Athens



University of Windsor

Course Discussion

Please Introduce Yourself:

- Name
- Where did you live before coming to Windsor/Canada?
- Fun fact – Interests
- What do you want to learn from this course?
- What are your goals?

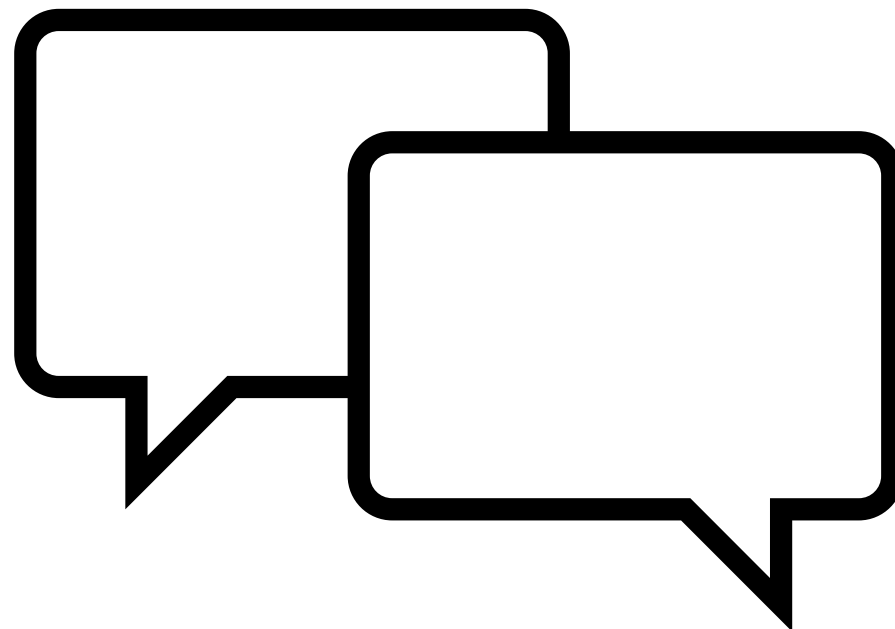


Course Discussion

- Help Each Other
 - Please share a learning resource or study tip.



● Let's Talk



Ask me
Anything



● Let's Talk

○ Your question does not have to be specific or well-defined:

- I need help with this.
- I am not clear on this.
- Could you go over this again?
- I am not sure what to do next.



Part I

Contact Information and Communication



In-person Meeting Etiquette

- Verify your schedule. Leave home early.
- Arrive on time, meaning before the meeting begins (we will end on time). Delayed arrivals severely disrupt the lecture delivery.
- Avoid multitasking. Pay attention to the lecture.
- Avoid leaving the classroom and returning while the lecture is on. We will be having breaks.
- Don't use your phone. Your phones should be in your bags, in silence mode.
- Don't disrupt speakers. Don't talk to each other.
- Actively participate.
- Take notes.
- No eating when during the meeting. Water/coffee are ok.



Virtual Meeting Etiquette

- Test your equipment early
- Arrive on time, meaning before the meeting begins (we will end on time)
- Avoid multitasking
- Notify before leaving
- Start muted, but with camera on, when logging in
- Don't disrupt speakers. Mute when not speaking
- Take notes
- Turn you Mic and Camera on when participating, look at the camera and speak clearly
- No eating when during the meeting. Drinks are ok
- The lecture will be recorded



Contact information & Communication

Dr. Andreas Maniatis

Email : Andreas.Maniatis@UWindsor.ca

- Email to request an appointment outside of office hours. Emails will be responded to within 24 – 48 hours during workdays.
- When contacting me:
 - You **must** use your **University of Windsor** email.
 - Please include your name and student number at the bottom of the message

Current information will be posted in the Course Information section of the Brightspace course.



Email

- It is **VERY** important that you check your university email regularly
- This is the only method I have for communicating with you
 - I will notify you of changes to the course, due dates, test dates, etc. through email
- When emailing your instructor, you **MUST** use your UWindsor email account
 - I will not respond to any email that comes from your personal email account.
 - When emailing, ALWAYS include your full name, section and student number at the bottom of the email

For example:

Peter Parker
COMP-2560-1
#123456789



Peter



Email Signature

- Even better, spend some time to create your personalized email signature: <https://www.uwindsor.ca/logo/tools-templates>

For example:



Dr. Andreas Maniatis

Instructor

Master of Management – Business Data Analytics

Andreas.Maniatis@UWindsor.ca

Tel.: (519) 567-5165

UNIVERSITY OF WINDSOR – ODETTE SCHOOL OF BUSINESS

401 Sunset Ave. · Windsor · Ontario · N9B 3P4



- Add your photo in Brightspace



Email

- Include a subject line:
 - This tells me what the email concerns

Example Subject Line: **Assignment 2 - Problem with Title Not Displaying Properly**



- If this is concerning a problem, detail what the problem is in the email **and** attach any supporting document.



Email Etiquette



KEEP THE CONTENTS OF YOUR EMAIL
PROFESSIONAL



PROOFREAD BEFORE SENDING



REMEMBER, THIS IS NOT A TEXT
MESSAGE TO A FRIEND. NO 'UR' OR
EMOJIS

Part II

Orientation and Agreement

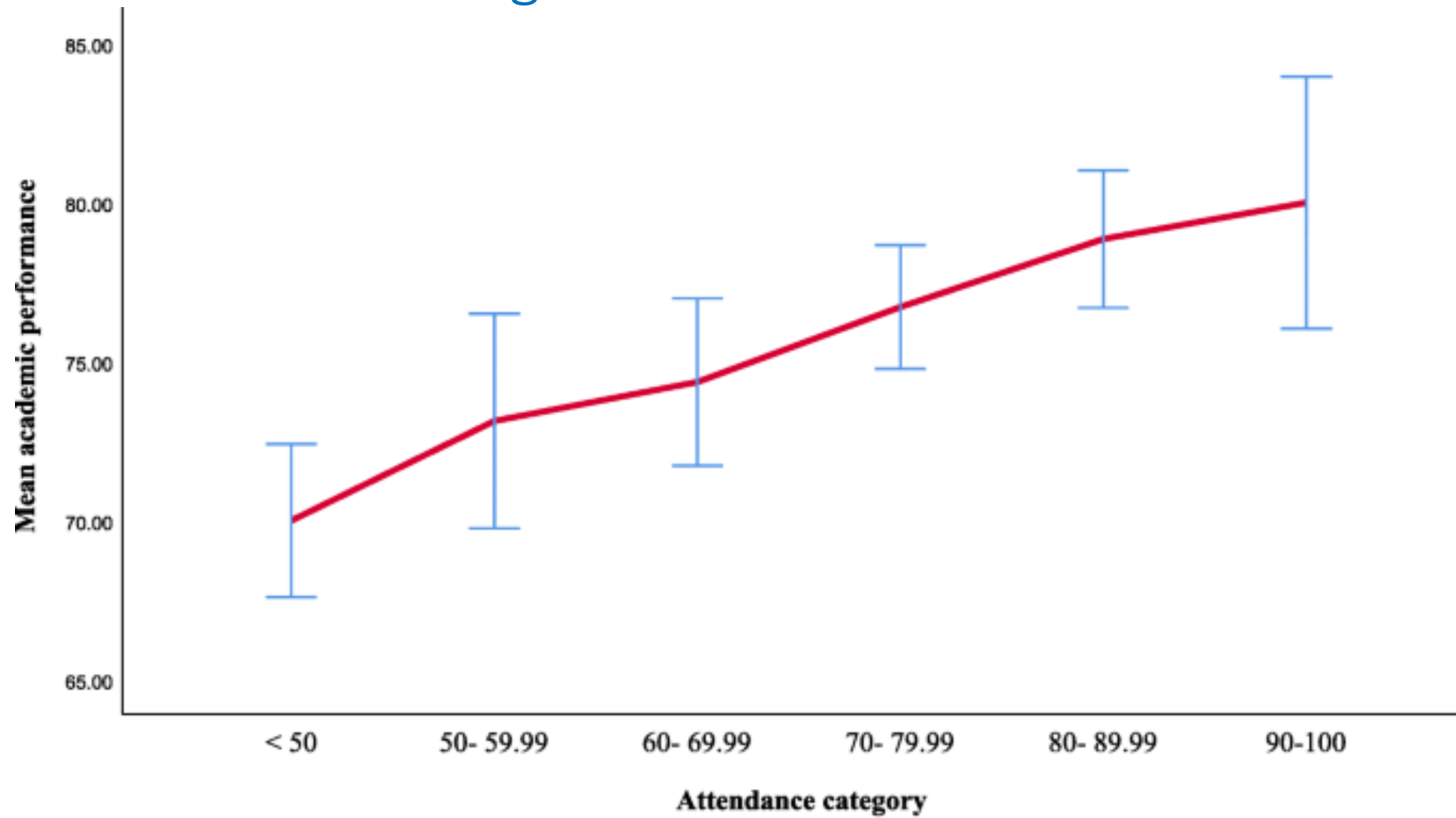


Orientation and Agreement

- **Prioritize** and **Focus** on your studies.
- Attendance: **Strongly Suggested!** Be focused on the lecture, don't be distracted (**don't use your phone**).



Orientation and Agreement



Orientation and Agreement

- **Prioritize** and **Focus** on your studies
- Attendance: **Strongly Suggested!** Be focused on the lecture, don't be distracted (**don't use your phone**)
- Please, be punctual
- Absence: Notify as early as possible
- Familiarize yourself with the available tools (Brightspace, etc.)
- Familiarize yourself with the University of Windsor Services (Student Support, Centre for Academic Excellence, etc.)
- Maintain a steady attending, learning and reading pace
- Maintain English as a working language
- Maintain business and personal ethics



● The four agreements

- • Be Impeccable with Your Work
- Always Do Your Best
- Don't Make Assumptions
- Don't Take Anything Personally

Be resilient!



Learning



From your instructor.



On your own.



From each other.

Share something interesting you've learned, such as

- News items.
- Learning resources.

Participate in live sessions and labs.

Form study groups.



Guided Learning



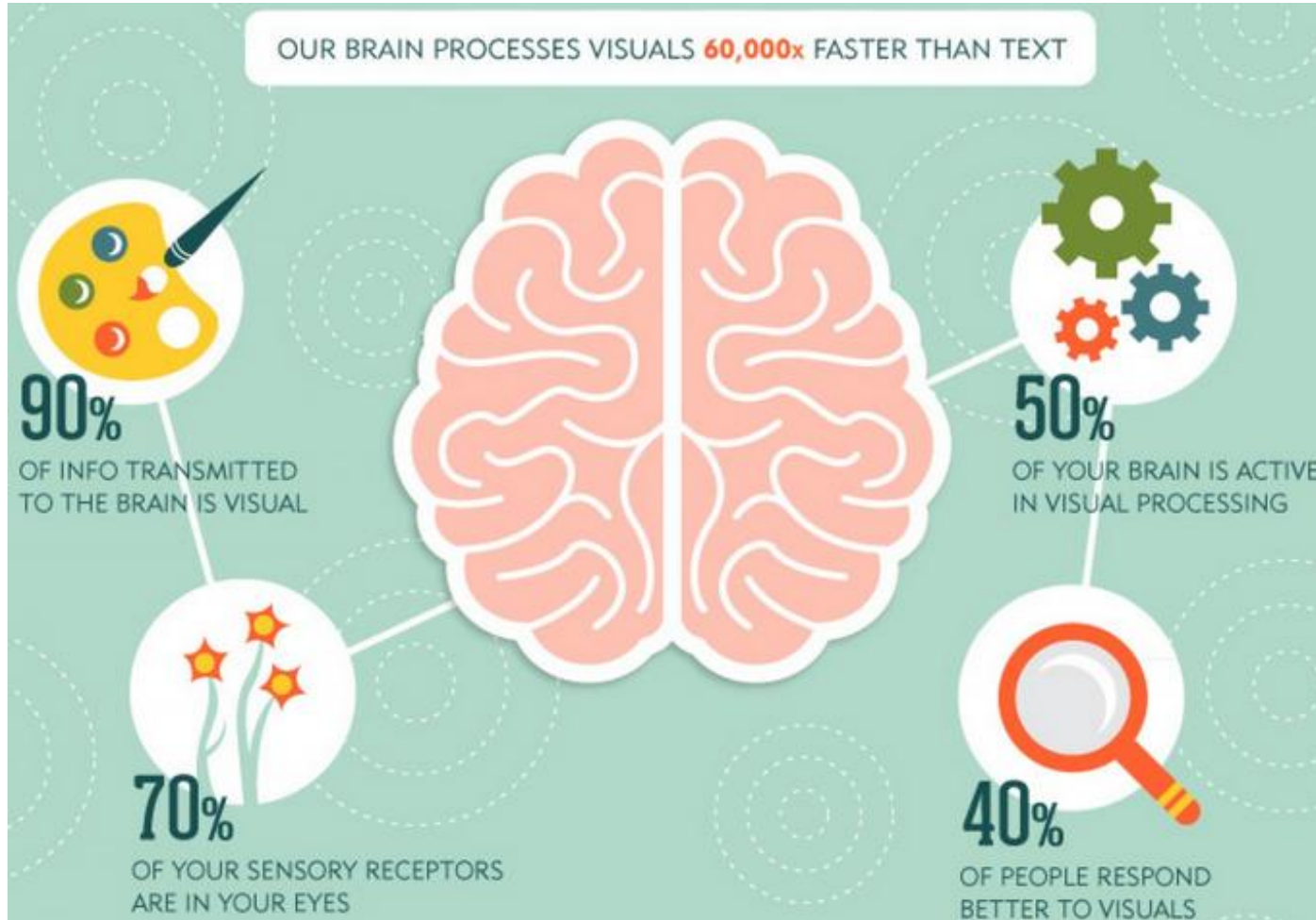
● The importance of senses

- Seeing 83 percent
- Hearing 11 percent
- Touching 3½ percent
- Smelling 1½ percent
- Tasting 1 percent
- 6th Sense ... ?

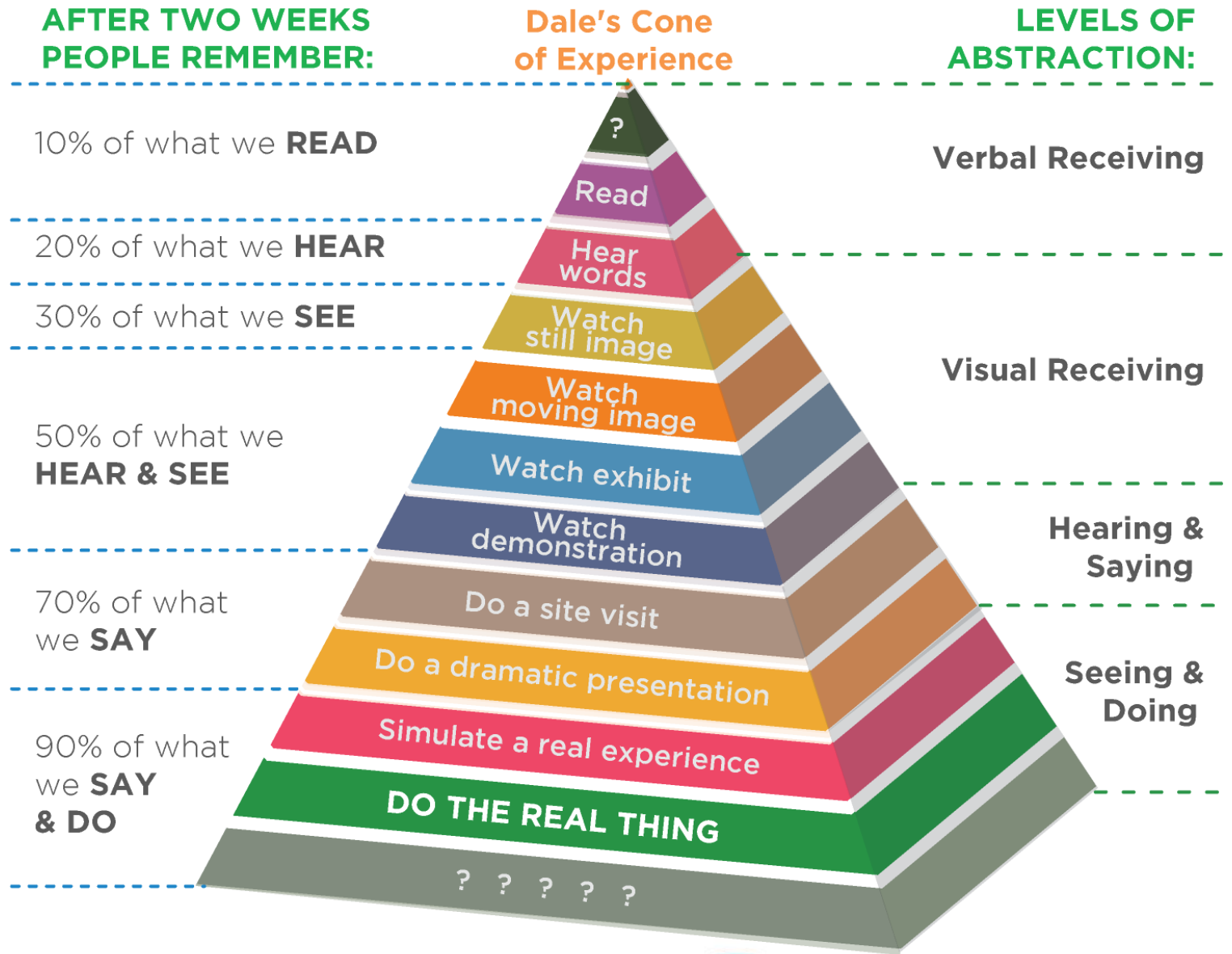
Source: Oklahoma State University
http://www.oces.okstate.edu/washita/uploaded_files/4h_Learning_Styles.doc



Biology in Visual Perception



Cone of Experience



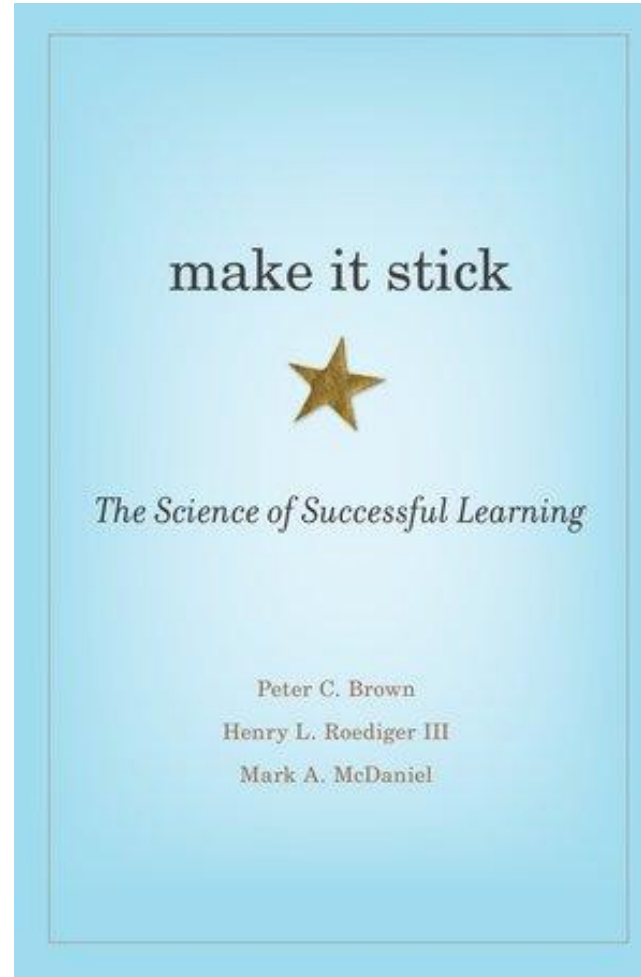
Source: [Learning Pyramid – Wikipedia](#)

(Edgar Dale's "Cone of Experience")



University of Windsor

Make it stick



● Make it stick

- Get organized to help you build mental models of the material
- **Repetition** to reinforce the concepts by forcing yourself to recall material
- Relate the new concepts to existing concepts you know
- Apply what you learned to solve problems.





Think about how you learn



- What is your personal learning style?



Keeping Notes: Methodology

- **Keeping notes**, either when attending a lecture and/or when reading, is a game-changer for your comprehension.
- **Preview the topic** before class so new ideas have a place to “stick.”
- **Focus on the professor’s cues**—repeated points, emphasis, and slide transitions.
- **Write down concepts, not verbatim sentences**; aim for understanding over transcription.
- **Organize notes visually** with headings, bullets, and spacing to show relationships.
- **Flag anything confusing** with a quick “?” or highlight to ask about later.
- **Capture examples and problem steps**—they’re often test material.
- **Review and tidy your notes the same day** to lock in learning.
- **SUGGESTION:** Keep your Notes **HANDWTITTEN!**



Keeping Notes: Tools

- **Handwritten notes:** The most effective to support cognition.
- **Software tools:**
 - **Obsidian** – Markdown-based “second brain” with backlinking and knowledge graph views (great for research).
 - **Joplin** – Open-source, privacy-focused note app with offline syncing.
 - **Logseq** – Open-source for task and knowledge management.
 - **Notion** – All-in-one workspace for notes, databases, to-dos, collaboration, and AI summaries. Ideal for organizing class notes and projects.
 - A complete list can be found here: <https://myndmess.org/>
 - Other:
 - Microsoft OneNote – Flexible digital notebook with sections, free-form pages, stylus support, audio recording, and searchable handwriting.
 - Google Keep – Simple, quick notes with color-coding, reminders, voice input, and OCR scanning.
 - Notability – Popular on Apple devices for combining typed, handwritten, audio, and sketch notes.



Finding the Answer

- I may not answer your question directly.
- Instead, I may guide you in finding the answer on your own.



The Learning Experience

- There are no shortcuts to learning.
- Mistakes are okay; just learn from them.
- We challenge you in the course so that you are well-prepared for life after the course.
- It is all about resilience.



Advice

Self-care (eat,
sleep, exercise).

Show up and do
the work.

Get organized.

Manage your time.

Put consistent
effort in every
week – DO NOT
CRAM!!!!!!

Read up on the
topic and build
your expertise.

Master the topic –
the grade will
follow.

Have fun.



Procrastination

- Due to the fact the most students spend many hours per week either working a part-time job, gaming and accessing social media sites, often little time is left for schoolwork.
- This program is a 'full-time' program requiring more than 20 hours per week of scheduled class time and between 12 – 20 hours of homework (depending on the student).
- “Putting off schoolwork, skipping assignments or class and leaving assignments until the very last minute, often leads to stress and often ends up biting us back in the long run.” ~Nikita
- Please, try to stay on top of your schoolwork this semester.



Getting Organized



What productivity tools do you use?



How do you manage your time?



How do you plan to manage the competing demands for your time?



How do you organize your notes?



What's expected of you



Review weekly content on your own.



Complete assignments individually or in teams, as indicated.



Contribute positively to your team for the mini group projects.



Part III

COMP.2800: Software Development:
Course and Competencies



University of Windsor

An Introduction to Advanced Programming

Yes, this is for you: A Course Essential for
**Advanced Software Development
Techniques!**



Quick Poll about Java

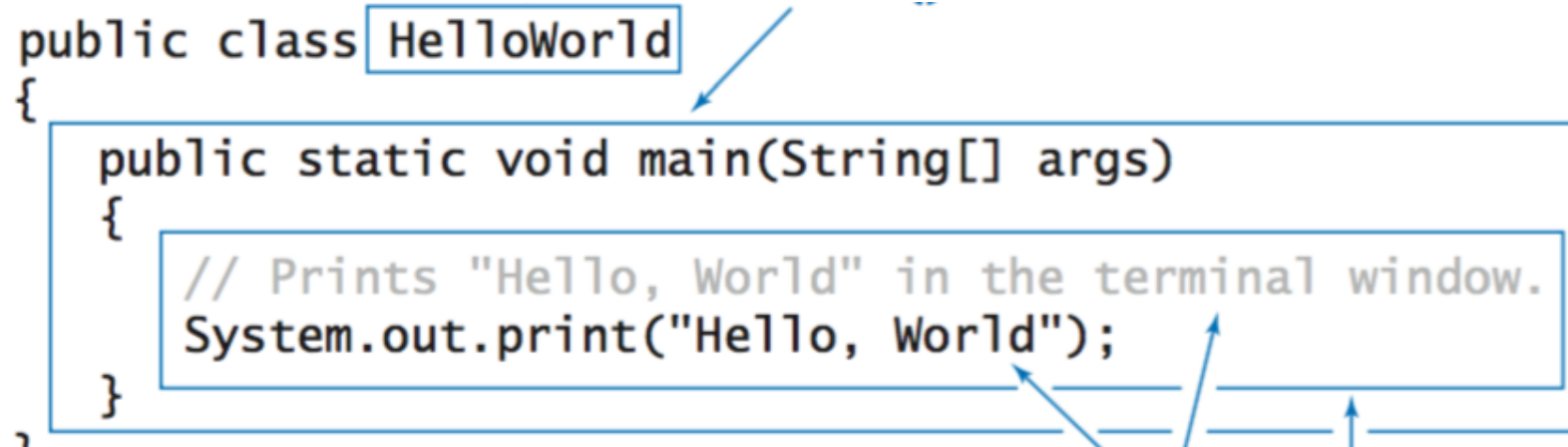
- Be sincere!
- Use the following Menti:
<https://www.menti.com/bly2d24wipy9>



Check your understanding

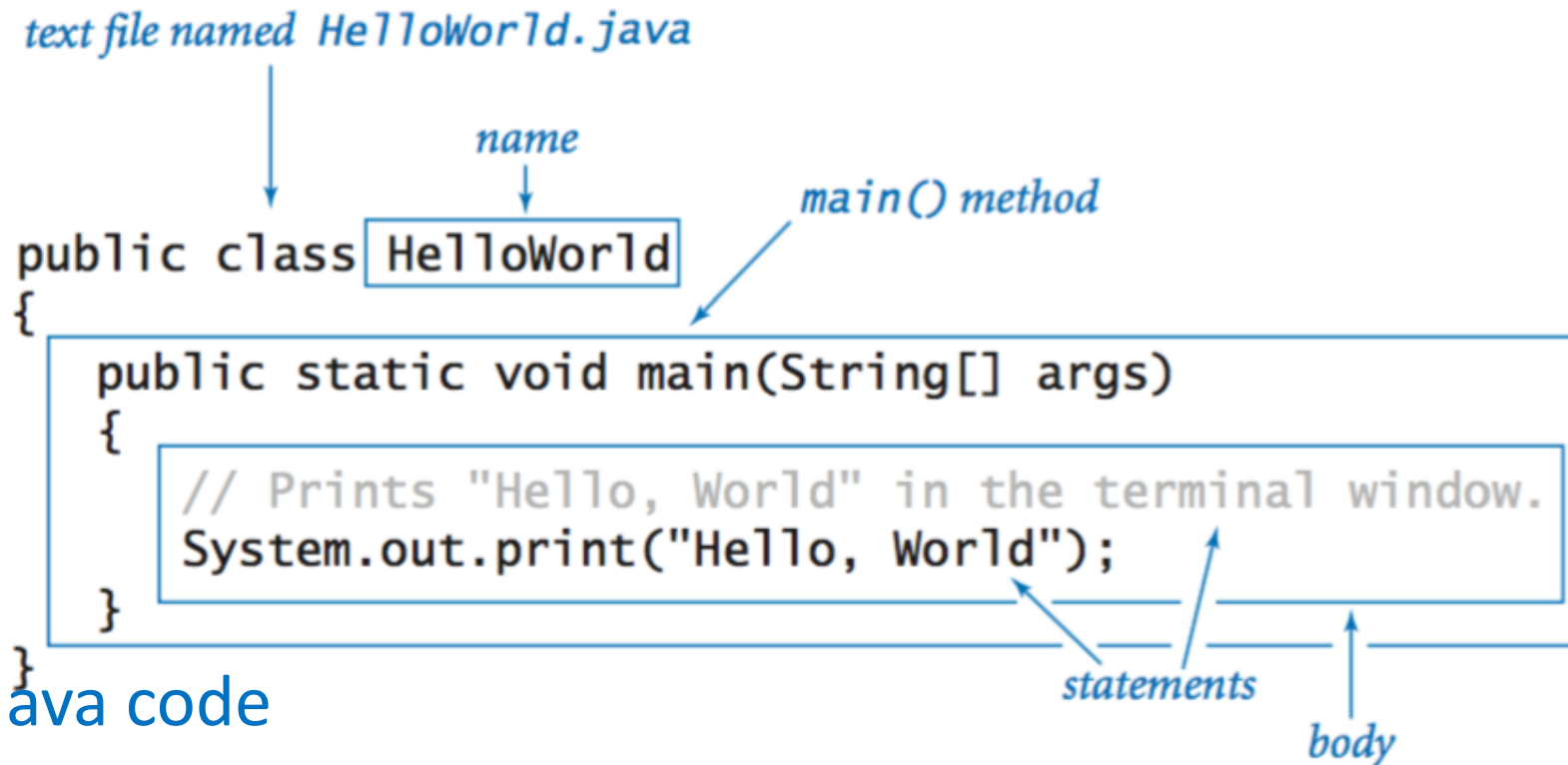
- What do you make of the image below?

```
public class HelloWorld
{
    public static void main(String[] args)
    {
        // Prints "Hello, World" in the terminal window.
        System.out.print("Hello, World");
    }
}
```



Check your understanding

- What do you make of the image below?



- Java code



● Check your advanced knowledge about Java



- Can we use Java to:

- ... access and retrieve data from a database?



- ... write data in a database?



- ... design and implement a fancy UI?



- ... call functions from other programming languages (i.e., C++)?

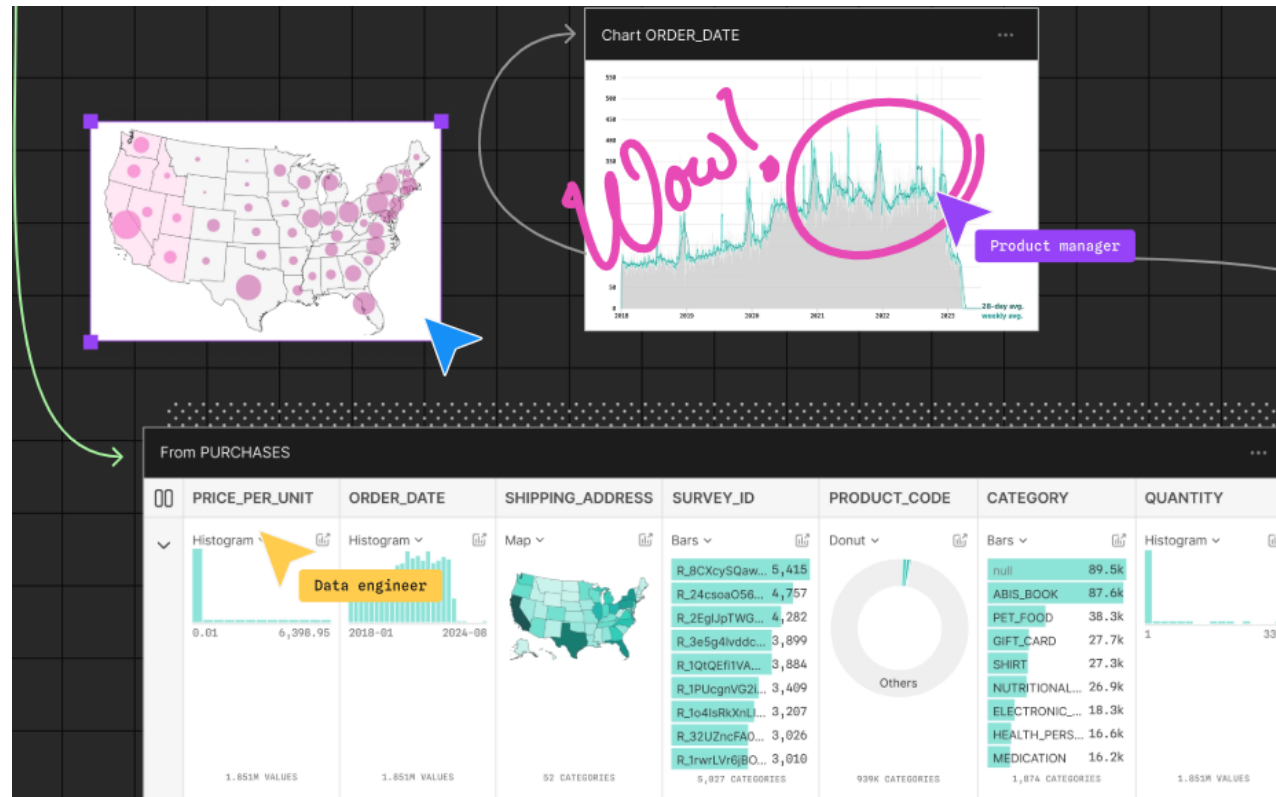


- Coffee is made of beans. Do we have Beans in Java?



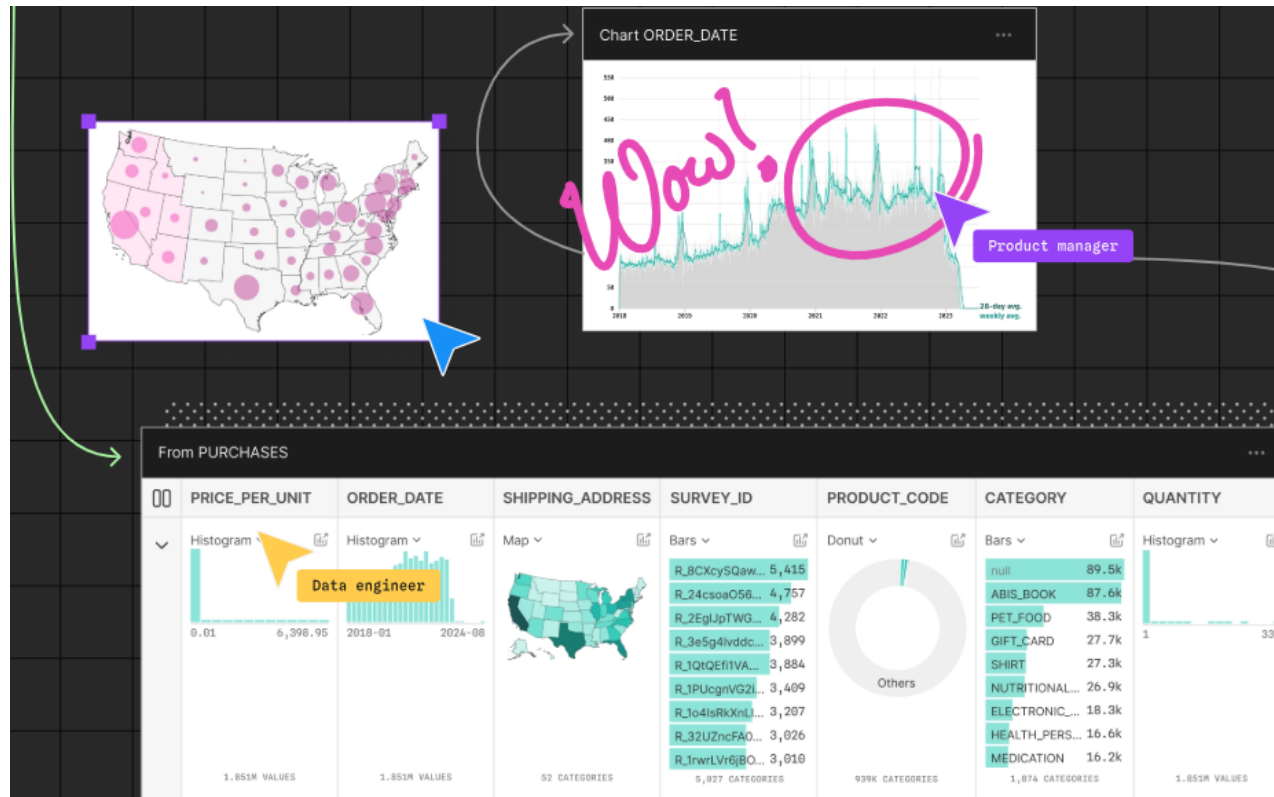
Check your understanding

- Can you create the dashboard below using Java?



Check your understanding

- Can you create the dashboard below using Java?



- YES! D3 visualization in Java (Other libraries include JavaFX and Swing)

About the course

- **COMP.2800: Software development**

- This course:
 - **Advances** students' **Programming Skills**, and
 - Introduces basic **Software Engineering** concepts and techniques through practice.
- Topics include, among others:
 - Event-driven programming,
 - Advanced GUI design,
 - Concurrent programming techniques,
 - Reflection and inter-process communication, etc.
- Basic software engineering topics include:
 - Requirements analysis and design,
 - Medium sized project management, and
 - Automated software engineering tool development.
- Prerequisites: COMP.2120 – Object-Oriented Programming Using Java



About the course: Learning Outcomes

- **COMP.2800: Learning objectives**
 - Develop programs using the event-driven paradigm
 - Develop concurrent systems
 - Design graphical user interfaces
 - Develop a correct implementation to satisfy a given set of software requirements
 - Demonstrate skills to become current with the advancement of programming languages
 - Retrieve useful information from Java API
 - Find a suitable design and implementation solution to a given problem using methods available from Java library



Part III

Lecture schedule



Lecture Schedule

- Course plan for Winter 2026 semester:
 - Duration: 14 weeks (last day of lectures: April 6th)
 - Two lectures per week
 - One lab per week
 - See Brightspace for weekly schedule
- Delivery mode: IN-PERSON
- Delivery format: Tutorial Course
[Read ALL course materials, review the posted lectures and complete assignments and be at the LAB to be marked]
- Educational session (80 minutes, 5 minutes bio-break)
- Office hours: Tuesdays and Thursdays, 10:00 – 12:00
 - MS Teams / LT-8106
 - TAs/GAs Office Hours: TBA



Part IV

Required tools, Equipment, and Learning Resources

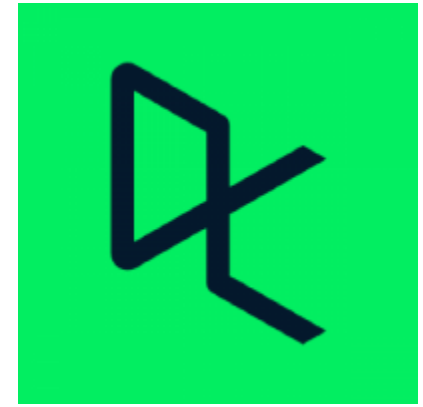


Required tools, Equipment, and Learning Resources

- Textbook:
 - *Core Java , Volume II-Advanced Features, 14th Edition*
Cay S. Horstmann
 - Available through Brightspace
- Supplemental tutorials (free):
 - *GeeksforGeeks*
 - Official Documentation (<https://www.oracle.com/java/technologies/jdtt-jsp.html>)
 - *DataCamp* (www.DataCamp.com)
- Open-source tools and technologies:
 - *Java*
 - *JDK / JRE*
 - *IDEs (various)*
 - *Google Colab*
 - *Others ... (Java libraries, such as D3.js, etc.)*
- A variety of case studies, videos, examples, and tutorials



DataCamp



- About DataCamp (www.DataCamp.com)
- Learn Data Science & AI from the comfort of your browser, at your own pace with video tutorials & coding challenges on Java, Python, R, Statistics & more
- Register as soon as you receive the invitation in your UWindsor email address
- DataCamp Assignments: *TBA*
- Issue and save the respective certificate(s)



Part V

Assessments



● Assessment

Assessments	%
Assignments (4)	20
Labs (10)	10
Mid-term examination (project)	15
Group project	25
Final examination	30
Total	100



Assessment: Assignments

- Up to four (4) Assignments
- Demonstrate the ability to apply methods and techniques to solve advanced coding problems using Java
- Explain the steps followed to solve a problem
- Assessments must be completed in person during lab sessions, and will be heavily based on in-class content/discussion, presented during lectures and labs

When writing lab tests you must abide by University of Windsor policies governing plagiarism and academic integrity. Test submissions may be subjected to review by automated tools to verify their originality.



Assessment: Labs

- Ten (10) Lab Assessments
- Demonstrate the ability to apply methods and techniques on advanced coding problems using Java
- Explain the steps followed to solve a problem
- Assessments must be completed in person during lab sessions, and will be heavily based on in-class content/discussion, presented during lectures and labs

When writing lab tests you must abide by University of Windsor policies governing plagiarism and academic integrity. Test submissions may be subjected to review by automated tools to verify their originality.



Assessment: Midterm examination

- In-person / in-class
- Individual Coding Task
 - Solve the problem using coding in Python
- Material covered till the date of the exam

When writing lab tests you must abide by University of Windsor policies governing plagiarism and academic integrity. Test submissions may be subjected to review by automated tools to verify their originality.



Assessment: Project

- Group Project
- Define a problem to solve
- Solve the problem using coding in Java
- Present the solution

When writing lab tests you must abide by University of Windsor policies governing plagiarism and academic integrity. Test submissions may be subjected to review by automated tools to verify their originality.



Assessment: Final examination

- In-person / in-class
- Format (TBV):
 - Part A
 - True / false questions
 - Multiple choice
 - Part B
 - Java Coding
- Material covered till the date of the exam

When writing lab tests you must abide by University of Windsor policies governing plagiarism and academic integrity. Test submissions may be subjected to review by automated tools to verify their originality.



Assignments

- **Academic Integrity**

You are required to do your own work!

- All assignments will be compared – any instances where assignments match will be handled as per the Code of Student Rights and Responsibilities:
[Code of Student Rights & Responsibilities](#)
- Plagiarism resources:
 - Tutorial: [Prevent Plagiarism | Academic Integrity](#)
 - Contract Cheating: [Contract Cheating | Academic Integrity](#)

[Academic Integrity Policy](#)



University of Windsor

How to be successful in this course

- **Do the work**

- We will have dedicated lab time for you to complete the work – don't avoid these times, don't take the work home and don't leave the work until the last minute. I am there to answer questions and provide help. We cover a lot of material – if you miss a class, you may miss an entire topic
- Some of the work will look simple but being able to master the technique may take longer than expected. If you leave it to the last minute, you will not get the work completed

- **Attend lectures**

- In addition to the assignments, we will be doing hands on exercises every class. If you miss the class, you may miss some important information that is not covered in the weekly assignments. Plus, you can now ...
- **Ask questions**
- If you do not understand a topic – ask. It can be in class, in lab, during office hours or via Teams.
- **Focus on your studies**
- Avoid distractions and make your studies your NUMBER ONE PRIORITY.



How to be successful in this course

- **Use the course lecture materials**

- If you need to further understand something you are doing – review the course materials.
 - Review the lecture notes and class exercises
 - Review any of the posted links in Canvas
 - Review your textbook
- DO NOT GOOGLE AN ANSWER – there are several bad answers out there – everything you need to know is contained in the provided lecture materials!
- Any answers that implement anything that we have not covered in class will automatically receive a mark of 0 for that question.
You will not be asked to implement anything that we have not covered in class.



Part VI

Important dates



Important dates

Date	Exam / assignment
~Weekly	Labs
~Bi-Weekly	Assignments
Wednesday, February 25 th , 2026 (TBV)	Mid-term Examination, in-person
Week 12	Group Project
April 9 th – 20 th , 2026 [TBA]	Final Examination, in-person
Note: Final exams will take place during the university specified final exam period. Students are advised to read Senate Policy on the Conduct of Tests and Exams.	



Thank you!
Questions?

Dr. Andreas S. Maniatis

Assistant Professor

Andreas.Maniatis@UWindsor.ca

<http://www.linkedin.com/in/andreasmaniatis>



University of Windsor